FAST NUCES — Karachi Campus

AI Project Report

Real-Time Sign Language Alphabet and Number Recognition Using Computer Vision and Machine Learning

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1. Executive Summary

Overview

This project presents the development of a real-time sign language recognition system. Using a standard webcam, the system successfully captures hand gestures and interprets them as letters or numbers from the sign language alphabet. The major goal is to make communication more accessible for individuals who rely on sign language in their daily lives.

Key Features

- Real-time recognition of hand gestures using a webcam.
- Automated collection and organized storage of gesture images across labeled folders.
- Image preprocessing to ensure consistency in size and quality across the dataset.
- A neural network-based classifier to identify and label hand gestures.
- Smoothing techniques applied to improve the stability of real-time predictions.
- Word and sentence construction from sequential gesture inputs.
- Text-to-speech output to vocalize recognized words and sentences.
- A user-friendly interface that allows manual adjustments and control over the system.

Core AI Techniques

- Computer vision for hand detection and gesture capture.
- Image processing to standardize input quality and dimensions.
- A Convolutional Neural Network (CNN) for gesture classification.
- Structured data collection and storage pipelines for dataset management.

2. Background

Sign language is the primary mode of communication for many individuals who are deaf or hard of hearing. However, because most people are unfamiliar with it, communication between sign language users and the general public can be quite difficult. This project aims to bridge that gap by developing a system capable of recognizing sign language gestures in real time and translating them into readable and audible output.

Project Objectives

- Develop a system that recognizes sign language gestures in real time.
- Use a webcam to capture hand gestures and determine their corresponding letters or numbers.
- Construct words and sentences from sequential gesture inputs.

- Provide audio output of the recognized text using text-to-speech.
- Demonstrate the system's potential as a practical communication aid.

3. Project Description & Scope

Scope

The system is designed to recognize individual letters (A–Z) and numbers (0–9) expressed through hand gestures. It does not currently support full sentence gestures or signs that require the use of both hands simultaneously.

Technical Overview

Category	Details
Language	Python
Computer Vision	OpenCV, CVZone
Machine Learning	TensorFlow / Keras (CNN)
Support Libraries	NumPy, Math, Time
Input Device	Webcam (real time)
Dataset	Custom — 36 Classes (A–Z, 0–9)
Model File	Pretrained CNN (.h5)

Libraries Used

1. OpenCV — Image processing and video capture
2. CVZone — Hand tracking and detection
3. TensorFlow / Keras — Model training and prediction
4. NumPy, Math, Time — Supporting utility operations

4. Methodology

Approach

The system was developed in a modular fashion, with each component built and tested independently before being integrated into the full pipeline. Gesture images were preprocessed and resized to a uniform dimension to ensure consistent model input. The CNN model was then

trained on this dataset and evaluated through real-time testing. Stability and smoothness were prioritized during the recognition phase to produce reliable and usable output.

5. Project Implementation

Structure

The system captures live video input through a webcam and detects the user's hand within each frame. The detected hand region is cropped and resized to match the model's expected input dimensions. The CNN then processes this input and predicts the corresponding letter or number. Recognized gestures are accumulated over time to form words and sentences.

Functionalities Developed

Data Collection Module

- Captures and stores images of hand gestures from the webcam feed.
- Supports both automatic and manual saving modes.
- Tracks and displays collection progress for each gesture class.

Dataset Management

- Automatically creates and organizes folders by gesture class.
- Provides options to review dataset statistics and delete existing data if needed.

Model Training Module

- Trains a CNN on the collected gesture dataset.
- Includes early stopping to prevent overfitting and unnecessary computation.

Real-Time Recognition Module

- Detects and classifies hand gestures from live webcam input.
- Applies prediction smoothing to reduce flickering and improve stability.
- Accumulates recognized characters to form words and sentences.
- Outputs the recognized text audibly using a text-to-speech engine.

Challenges Faced

- Variability in lighting conditions affected the consistency of hand detection.
- Inconsistent image quality across collected samples introduced noise into the dataset.
- Maintaining smooth and stable predictions in real time required additional post-processing.
- Running multiple processes concurrently, such as detection, classification, and audio output, required careful optimization.

6. Results

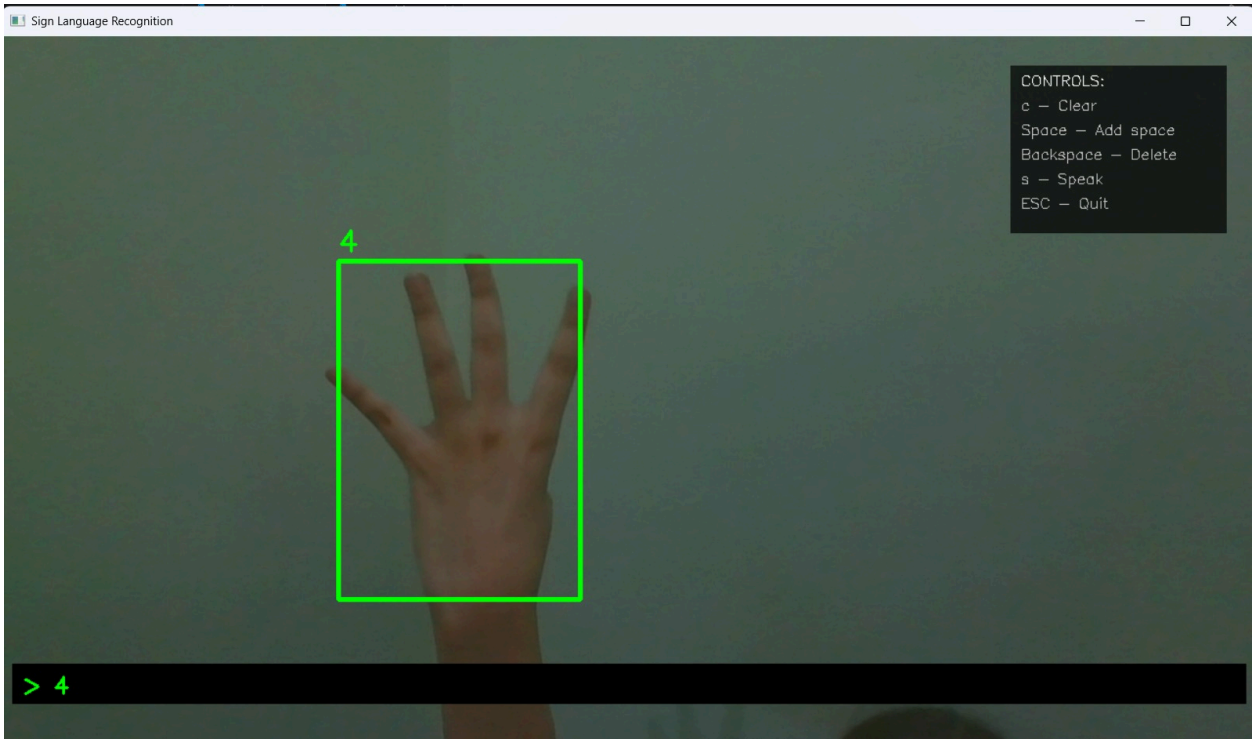
Project Outcomes

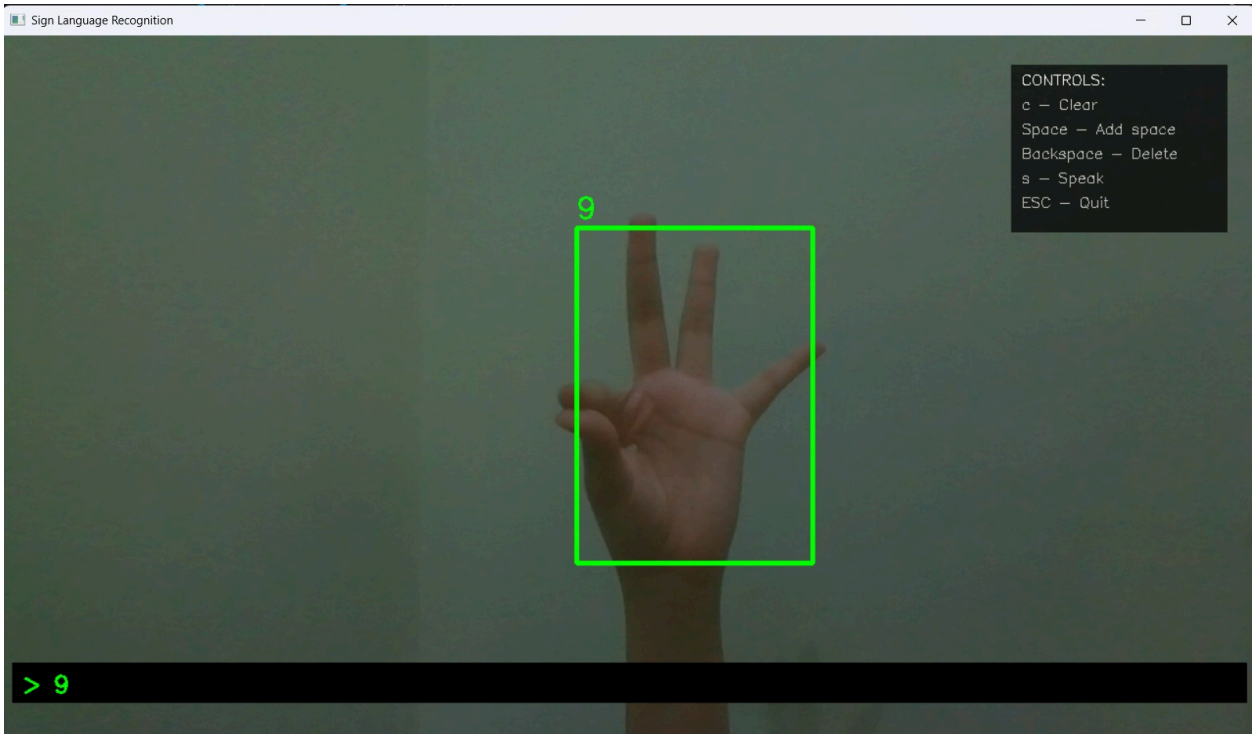
- The system successfully recognizes sign language letters and numbers in real time.
- Sequential gestures are accurately combined to form words and sentences.
- Text-to-speech functionality correctly vocalizes the recognized output.

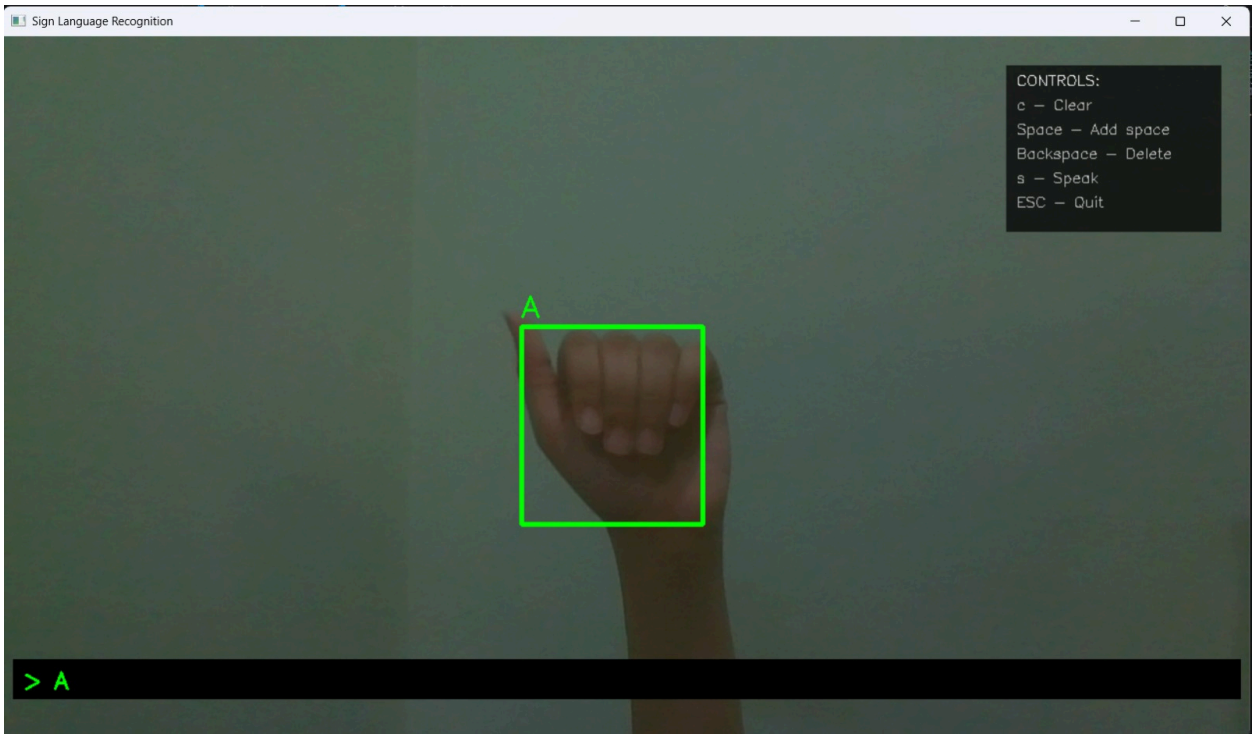
Validation

- The system was evaluated across a range of hand gestures and lighting conditions.
- Stability and smoothness of real-time predictions were verified through repeated testing.
- Recognition accuracy for all 36 classes (A–Z, 0–9) was confirmed during the validation phase.

Example Use-Cases:




Sign Language Recognition
CONTROLS:
c – Clear
Space – Add space
Backspace – Delete
s – Speak
ESC – Quit
9
> 9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